

HDR Photography

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RAW Photos

- Many modern cameras capture more than 8 bits per color channel
- Most screens and printers are only capable of displaying 8 bits per color channel, so data is often thrown away while processing or displaying the image
- The raw data from the camera contains much more information than a more typical image file, but the data can be hard to process because most companies have their own proprietary raw format

Contrast Stretching

- Converting a a fujifilm xtrans .RAF file to a 16 bit .ppm file produces some undesirable results
- Fujifilm xtrans sensors have a disproportionate amount of green sensors, and so the image looks like the grinch came for a visit
- Colors can be adjusted using a technique called contrast stretching, which I have spent the last several weeks trying to figure out
- Due to lack of access to fujifilm data and a small team size of 1, the results produced are less than desirable.

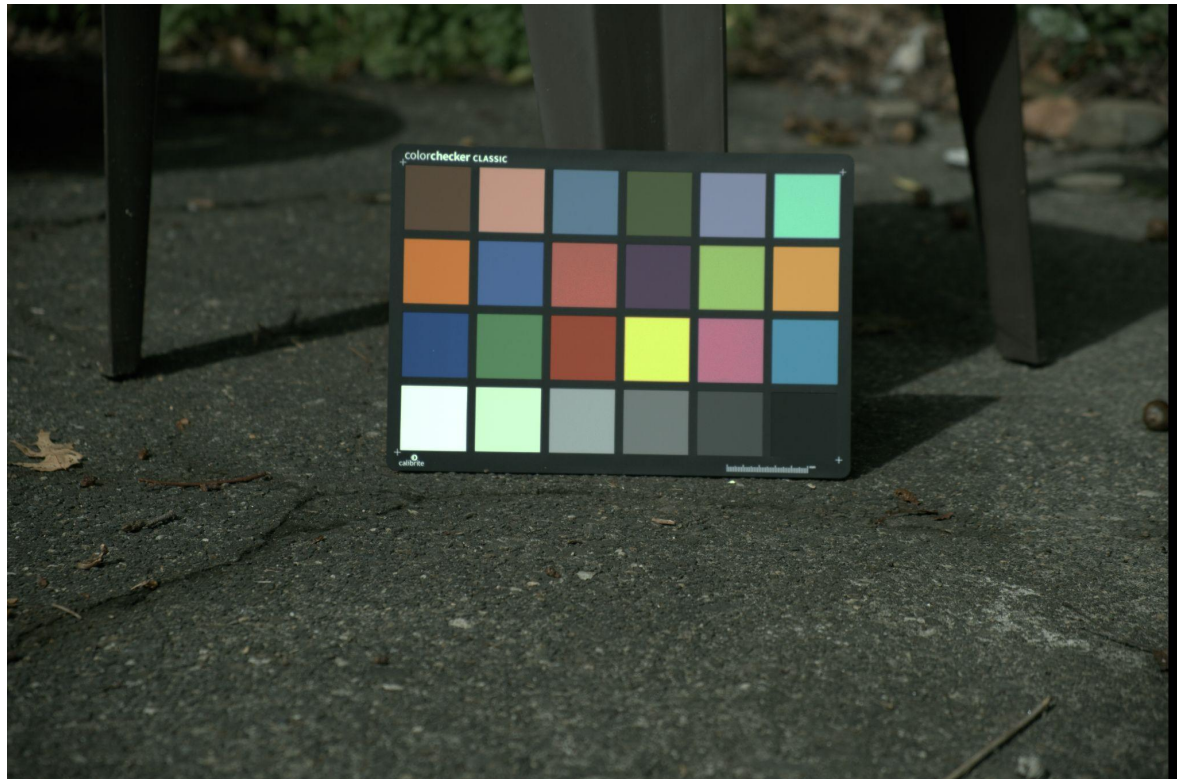
No Contrast Stretching (.ppm from .raf)



Linear Piecewise (.ppm from .raf)



Bilinear Piecewise (.ppm from .raf)



Affinity Photo (.jpg from .raf)



Apple Conversion (.jpg from .raf)

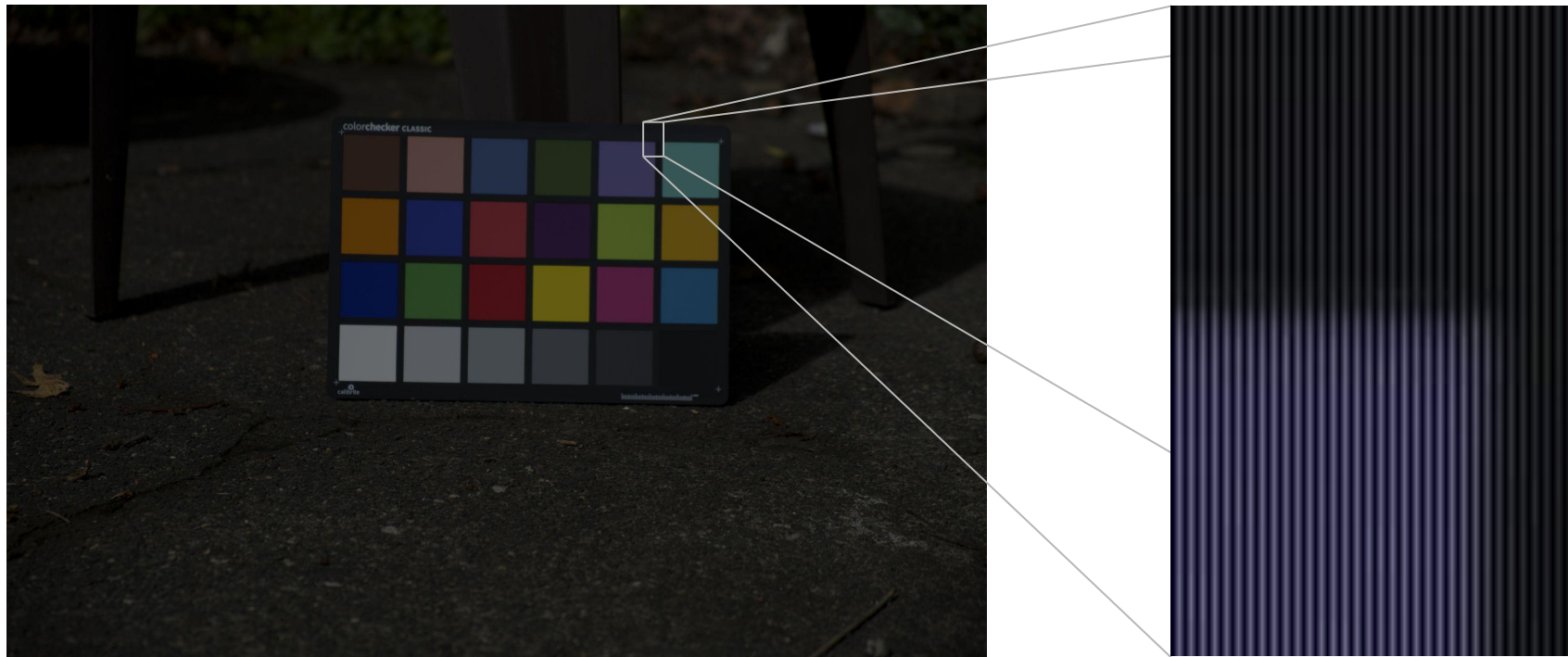


Over/Under Exposure

- Dynamic range is the difference between the lightest and darkest section of an image
- Sometimes the dynamic range of a scene is too great, and the camera ends up not capturing useful data in the light or dark section of the image
- Images can be stitched together to produce an image with detail in all parts of the image. The range of colors will not be preserved, but detail in the image is increased at the detriment of color accuracy



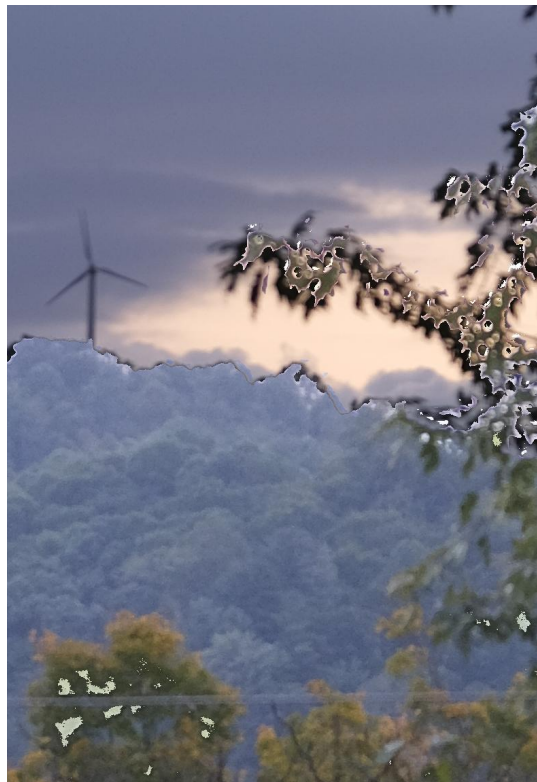
Pixel Array



Pixel Average



Pixel Select



Going Forward

- Use more complex algorithms - Weighted average, region selection
- Look into 16 bit color - Many cameras produce 16 bit color data, but it can be difficult to access
- Use a scriptable camera to take the images

Sources/External Code Used

dcraw - open source library written by Dave Coffin

Apple Preview

Affinity Photo

Mistakes

